

Project Initialization and Planning Phase

Date	15 March 2024
Team ID	
Project Title	Rising Waters
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	Develop an intelligent flood prediction system using Machine Learning to provide accurate early flood warnings and support disaster preparedness.
Scope	Analyze rainfall and weather data, train multiple ML models (Decision Tree, Random Forest, KNN, XGBoost), deploy the best model using Flask and IBM Cloud.
Problem Statement	
Description	Existing flood forecasting methods often provide delayed or inaccurate warnings, reducing preparedness and increasing losses.
Impact	Early prediction helps authorities plan evacuations, allocate resources efficiently, and reduce loss of life and property.
Proposed Solution	
Approach	Collect historical weather data, preprocess it, train ML models, select the best-performing XGBoost model, and integrate it into a Flask web application
Key Features	• Machine Learning-based flood prediction • Real-time prediction through Flask web app • XGBoost model with high accuracy • Dashboard for authorities • IBM Cloud deployment support

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Flood prediction dataset (Kaggle, CSV/Excel)